

EC2 Setup

I used an Amazon Linux EC2 instance in us-east-1 with 80 GiB storage. I installed Java 17 (Amazon Corretto), Python 3.9, and created a virtual environment with PySpark 4.0.2, Jupyter, pandas, pyarrow, and matplotlib. I used tmux to keep the session alive and accessed Jupyter through SSH port forwarding on port 8888.

Dataset

January 2026 NYC TLC trip records (yellow and green taxi), downloaded from the class S3 bucket *de300-hw3-nyctlc-549787090008-us-east-1-an*.

Data Cleaning

Starting with 3,765,161 combined rows, I applied six cleaning rules:

- Removed rows with missing pickup/dropoff timestamps
- Removed trips with non-positive distance
- Removed trips with negative fares
- Removed trips with negative total amount
- Removed trips where dropoff was before pickup
- Removed trips longer than 24 hours

This removed 208,124 rows (5.53%), leaving 3,557,037 clean records.

Analytics Results

1. Which taxi type had more trips?

Yellow: 3,518,128 trips. Green: 38,909 trips. Yellow taxis account for ~99% of all trips.

```
+-----+-----+
|taxi_type| count|
+-----+-----+
|   yellow|3518128|
|   green | 38909|
+-----+-----+
```

2. Average fare by taxi type?

Yellow: \$21.08. Green: \$16.00. Yellow taxis have higher average fares.

```
+-----+-----+
|taxi_type|avg_fare|
+-----+-----+
|   yellow|   21.08|
|   green |   16.0|
+-----+-----+
```

3. Average trip distance by taxi type?

Yellow: 6.76 miles. Green: 12.99 miles. Green taxis serve outer boroughs, resulting in longer trips despite lower fares.

taxi_type	avg_distance
yellow	6.76
green	12.99

4. What hour had the most pickups?

6 PM (hour 18) with 232,570 pickups - evening rush hour. The least busy hour was 4 AM with 27,792 pickups.

hour	count
18	232570
17	224277
15	216305
19	211118
20	208072
21	207428
16	206485
14	202317
13	190970
22	190251
12	183146
11	167923
10	158111
9	154087
23	148822
8	146020
7	112027
0	109013
1	75623
6	62649
2	52470
3	37491
5	32070
4	27792

5. Percentage of trips under 2 miles?

51.95% - over half of all taxi trips are short rides.

6. Top pickup/dropoff location pairs?

The top route was PULocationID 237 to DOLocationID 236 with 23,273 trips. The top 3 pairs all involved locations 236 and 237 (Upper East Side).

PULocationID	DOLocationID	count
237	236	23273
236	237	20188
236	236	16051
237	237	14815
161	237	9969
237	161	9155
142	239	8734
239	238	8523
161	236	8250
141	236	8217

only showing top 10 rows

7. Weekend vs weekday trips?

Weekday: 2,537,387 trips, avg distance 6.99 mi, avg fare \$21.36. Weekend: 1,019,650 trips, avg distance 6.42 mi, avg fare \$20.21. Weekdays have more trips, slightly longer distances, and higher fares.

day_type	trip_count	avg_distance	avg_fare	avg_total
weekend	1019650	6.42	20.21	28.09
weekday	2537387	6.99	21.36	30.23

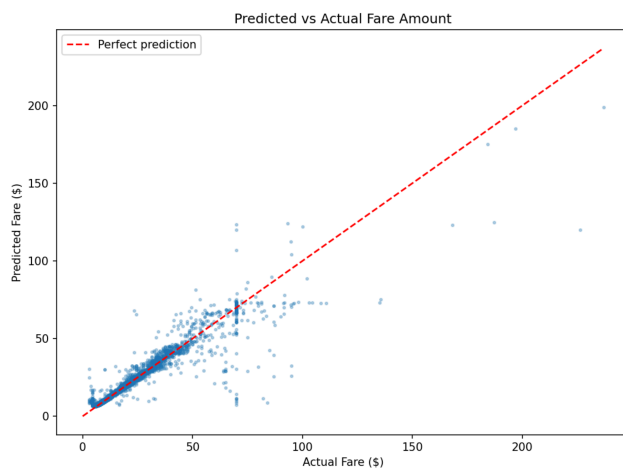
8. Fare prediction model.

I trained a Random Forest regressor (50 trees, max depth 8) using non-fare features: trip_distance, PULocationID, DOLocationID, passenger_count, hour, day of week, and trip duration in seconds. 80/20 train/test split.

Results: Train RMSE = 6.47, Test RMSE = 6.81. The close values suggest no overfitting.

Feature importance: trip_distance (0.50) and trip_seconds (0.39) dominated, accounting for ~90% of predictive power. This makes sense since NYC taxi fares are metered by distance and time. Location IDs contributed modestly (~10%), while passenger count, hour, and day of week were negligible.

```
Feature Importances:
trip_distance: 0.4955
PULocationID: 0.0657
DOLocationID: 0.0423
passenger_count: 0.0032
hour: 0.0025
dow: 0.0006
trip_seconds: 0.3901
```



S3 Output Paths

All results written to `s3://yegon-de300-hw3/nyc-taxi-assignment/`:

- trips_by_type.csv
- avg_fare_by_type.csv
- pickups_by_hour.csv
- fare_prediction_plot.png

AI Usage

Used Cursor AI to assist with brainstorming the PySpark pipeline structure, debugging S3 credential issues, and drafting small code snippets. All code was reviewed, adapted, and verified by running on EC2. Results were checked for correctness against expected patterns in NYC taxi data.